Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Bab 1

PENDAHULUAN

Bab 2

TINJAUAN PUSTAKA

[10pt,swedish, openany]book
[]graphicx []color alltt [T1]fontenc [utf8]inputenc
[top=100pt,bottom=100pt,left=68pt,right=66pt]geometry
lipsum
[swedish, english]babel
fancyhdr [C]5
titlesec

2.1em

0pt50pt2
float
[tableposition=top]caption
hyperref

Laboration report in Statistics

Laboration X

Course code

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liu_logo.png

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Chapter 3

Introduction

Introduce the laboration; the data set used, the overall goal of the exercises and add any preparatory work, for instance reading data sets and loading packages. Any output should be hidden unless specifically asked for.

«include = FALSE»=

@

Chapter 4

Exercises

Example of knitR

Example on how to run R-code directly in LaTeX

I run some code:

«echo=TRUE»= Some code library(xtable) Loads package xtable data(iris) mean(iris$Sepal.Length$)x <
$-10x$@

The output is a bit ugly so I hide some aspects of the code with:

«echo=FALSE»= Some code library(xtable) data(iris) x <- 10 x @

I write my report on the dataset iris.[**?**] I use references when needed. If I want to reference a specific page I can do that inside the citation. [**?**, p. 2-4] Still only one reference will be added to the bibliography.

I have read the data set iris which consists of nrow(iris) observations. The first observations can be seen here:

Showing output directly from R looks very ugly:

«echo=FALSE»= Print the data head(iris)

@

It is much nicer (and efficient) to use for example the package *xtable* to produce structured output.

«echo=FALSE, results='asis'»= Print a xtable table print(xtable(head(iris,n=20), caption="A table with data"), include.rownames = FALSE, Removes the row numbering table.placement = "H") Specifies the placement of the table, note that these two commands need to be placed inside the print() and not the xtable() @

If I use the caption argument in the xtable() function, a caption is automatically generated. This caption is automatically added to a list of tables (if that is used). This process is called floating in LaTeX and is done automatically by the xtable() function.

Figures

I have used my data, which can be seen on page 3, to create the nice looking figure 4.1 located on page 4.

Figures created in R are not automatically created in a floating environment (like tables), so this needs to be done manually in LaTeX with:

```
\begin{figure}
    \centering
    \includegraphics{} % R-code can be added here instead of the \includegraphics{}
    \caption{Caption}
    \label{fig:my_label}
\end{figure}
```

«echo=FALSE, fig.height = 4, fig.width = 7»=
with(data=iris,plot(Sepal.Length,Sepal.Width,main="My figure",col="Red", las = 1)) @

Figure 4.1: A figure

4.1 Exercise 1 - Linear regression

I have done an awesome analysis with linear regression on all nrow(iris) observations from the iris data set:

«echo=FALSE, results='asis'»= analys <- lm(Sepal.Length Sepal.Width, data = iris) xtable(analys, caption="Some output from the regression analysis") @

And a corresponding ANOVA table.

«echo=FALSE, results='asis'»= print.xtable(xtable(anova(analys), caption = "ANOVA")) @

If a list of tables was included in the document, all of these tables will be automatically listed there.